

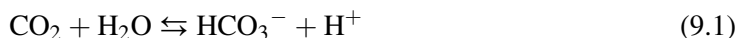
Mechanisms of action of carbonic anhydrase inhibitors: zinc binders

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9.1 Introduction

The reversible hydration of carbon dioxide is a minimal transformation which assumes paramount importance within our world of carbon-based life, operating in a water-based medium. All forms of life on Earth share the same biochemistry, which relies on the infinite number of chemical transformations on the “carbon” element. The conversion of carbon into its biologically fruitful form, as well as its detracting, is chemically represented in Eq. (9.1).



The kinetics of this reaction has been determined, and it has been proven to be significantly below the threshold for the biochemical transformations in maintaining the evolution of life. In this context are the carbonic anhydrase (CA; EC 4.2.1.1) enzymes, whose role is to speed up this equilibrium to match the biological needs. The abundance of genetic families and the multitude of isoforms contained within them clearly reflect the remarkable physio/pathological values of CAs. The features of exclusivity and selectivity applied to biological targets accurately fit with the medicinal chemistry recipe for success, and the CAs are typical examples. As a matter of fact, CAs are validated targets for the treatment of various diseases (most of which are discussed in this book). In addition, CAs have shown promising outcomes in the fields of anti-infectives, neglected diseases, and diseases affecting the central nervous system (CNS), such as neuropathic pain and memory, as well as in biotechnological applications [1–3]. The consequences of such enormous interest toward CAs were that many resources have been drained from various disciplines in order to determine the mechanistic and structural aspects of these enzymes.

9.2 Mechanistic insights into Zn-containing α - and β -CAs

As discussed in Chapter 2, the active site architecture in α -CAs was determined by means of X-ray crystallographic experiments and it appears conically shaped with a width of 12 Å and a depth of 13 Å. The Zn(II) ion, which is essential for the catalytic activity, is located at the bottom of such a cavity, coordinated in a tetrahedral fashion to three conserved His residues (His94, His96, and His119) and a water molecule/hydroxide ion (Fig. 9.1A) [4–6].

The metal-bound water molecule/hydroxide ion is involved in a network of hydrogen bonds which determine the enhancement of its nucleophilicity. In particular, it is hydrogen bonded with the hydroxyl moiety of the conserved Thr199 residue and with two water molecules, located on two opposite sides: the first, also called the “deep water,” is located in a hydrophobic cavity delimited by conserved residues in positions 121, 143, 198, and 209, while the second is in a hydrophilic environment toward the entrance of the active site (taking hCA II as a representative, Fig. 9.1B) [7]. Having defined the player (i.e., the metal-bound water molecule/hydroxide ion) for the CA catalyzed reaction to occur, the various stages involved are depicted in Fig. 9.2.

The catalytic cycle reported above begins with the carbon dioxide substrate allocated for catalysis within a hydrophobic enzymatic pocket (B), which is then converted to bicarbonate through the nucleophilic attack from the zinc-bound hydroxide ion, and (C) the

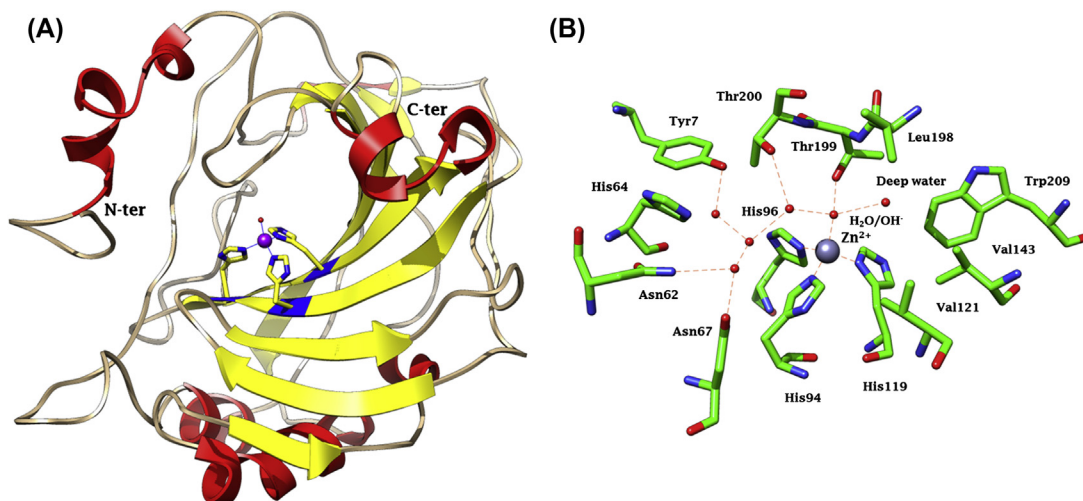


Figure 9.1

(A) Ribbon diagram of the hCA II structure (PDB accession code 1CA2), chosen as a representative CA isoform. The active site Zn(II) coordination is also shown. Helix and β -strand regions are colored red and yellow, respectively; (B) Zn(II) is tetrahedrally coordinated to the three catalytic His residues and a water molecule/hydroxide ion [3].

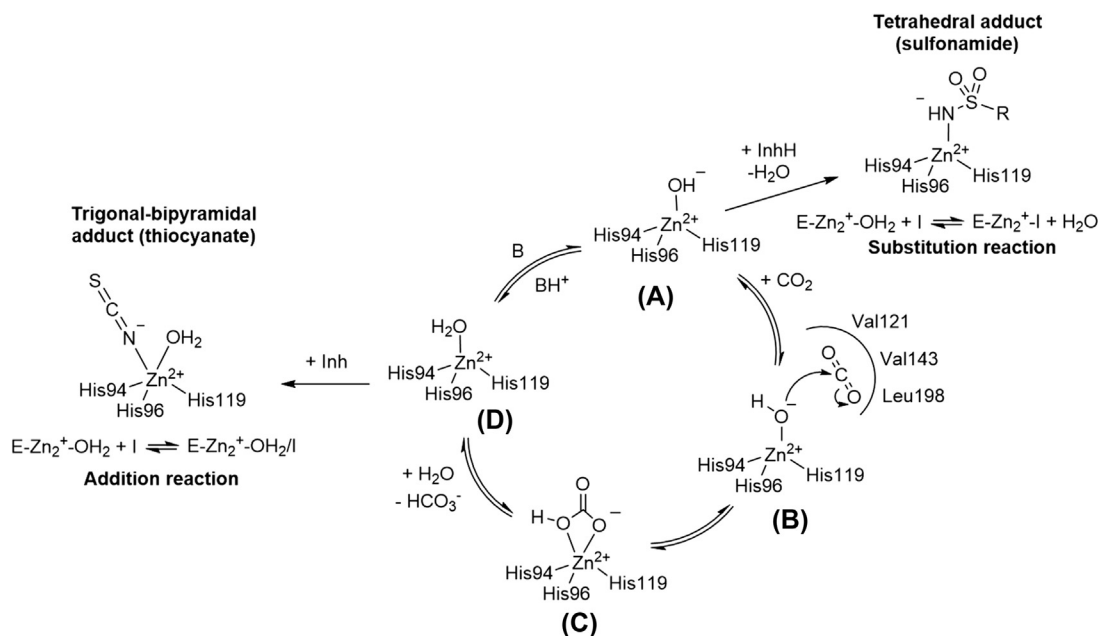


Figure 9.2

Catalytic cycle and inhibition mechanism of the α -CAs [1,7].

bicarbonate bidentately coordinated to the $\text{Zn}(\text{II})$ ion is displaced by a water molecule in such a way that (D) the enzyme with water coordinated to $\text{Zn}(\text{II})$ is generated. The D intermediate represents the inactive form of the CA enzyme, which is restored back to A (active form) upon a deprotonation assisted by an internal proton shuttling residue (e.g., His64 in isozymes I, II, IV, VI, VII, IX, and XII–XIV) or by buffers present in the medium [1,3,8,9]. This last step is also the rate-determining step for the entire cycle. Striking evidences on the reaction mechanism discussed above [1,3,8,9] have come from crystallographic studies which showed the entrapment of the CO_2 substrate within the CA II active site (i.e., adduct B in Fig. 9.2). As reported in Fig. 9.3A, CO_2 was bound within a hydrophobic small pocket delimited by Val121, Val143, Leu198, and Trp209 residues and properly oriented to undergo the nucleophilic attack by the zinc-bound hydroxide ion on its *sp*-carbon atom [10]. The other contribution is the hCA II/ HCO_3^- complex (i.e., adduct C in Fig. 9.2) [10]. As reported in Fig. 9.3B, the HCO_3^- molecule lies in the same plane defined by the CO_2 molecule and the $\text{Zn}(\text{II})$ -bound hydroxide ion, and is tetrahedrally coordinated to the catalytic metal ion through one of its oxygen atoms.

The elucidation of the hCA II/ CO_2 and hCA II/ HCO_3^- adducts allowed us to obtain clear evidence of the enzymatic reaction, but most importantly they provided solid structural data in support of new research and developmental studies in the field.

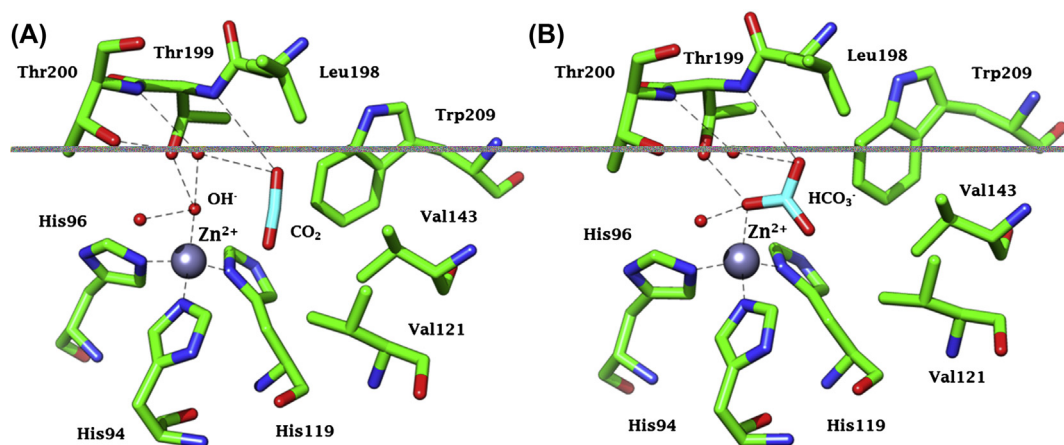


Figure 9.3

Active site of hCA II showing: (A) the position of the CO_2 molecule (PDB accession code 2VVA) and (B) the binding of the bicarbonate ion (PDB accession code 2VVB) [3].

For the purposes of this chapter, the authors intentionally skip additional aspects of the reaction mechanisms of α -CAs (i.e., the enhancement of the activity will be discussed in Chapter 22). The reader is advised to refer to the β -CA class, which presents the $\text{Zn}(\text{II})$ ion coordination sphere, which is quite different when compared both to the α -class and the β -family. For instance, the prokaryotic β -CAs have the $\text{Zn}(\text{II})$ ion coordinated by two Cys, one His, and a carboxylate moiety from an Asp residue (Fig. 9.4A), whereas in the chloroplast enzyme a water molecule replaces the Asp (Fig. 9.4B) [11–14].

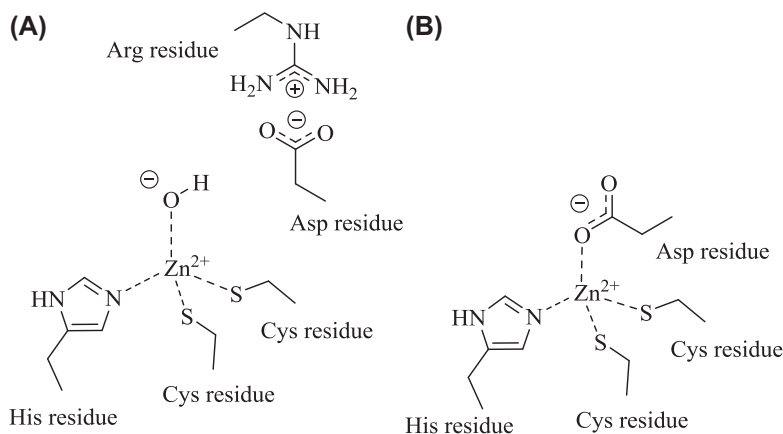


Figure 9.4

Schematic representation of the $\text{Zn}(\text{II})$ coordination sphere in β -CAs: (A) *P. purpureum* [11] and *E. coli* [12] enzymes; (B) *P. sativum* chloroplast and *M. thermoautotrophicum* enzyme [12,13] as determined by X-ray crystallography.

Since no water is directly coordinated to Zn(II) for some of the β -CAs, a different catalytic mechanism was proposed [15] and accounted for the active site being “blocked” at pH values of 7.5 or lower, when the carboxylate moiety of an Asp remains coordinated to the Zn(II) ion (as in Fig. 9.4A). However, at pH values over the threshold of 8.3, the Asp residue forms a salt bridge with a conserved Arg residue in all β -CAs, thus allowing a water molecule/hydroxide ion to coordinate the metal ion (Fig. 9.4B). The catalytic mechanism of this β -CA in the “opened” active conformation is then speculated to proceed in analogy with the α -class as in Fig. 9.2.

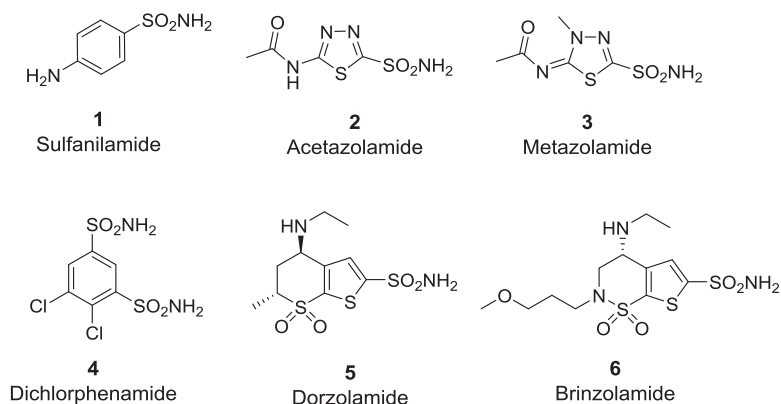
9.3 Zinc-binding moieties

The issue of developing specific CA modulators (i.e., inhibitors or activators) is particularly relevant as the administration of nonisoform selective drug-based CAIs has been proved to determine several side effects, and thus it narrows down the CAI-drug benefit/risk ratio [16]. In this context, many efforts have been made with the aim of tackling this problem, and two complementary approaches have been successfully developed: (1) the “tail” approach, which consists of decorating well-known molecular scaffolds with tail sections possessing the most variegate physicochemical properties; such a strategy allows us to make use of the variability, among the CA isoforms, of the amino acids composing the rim’s active sites; (2) conversely the “ring” approach focuses on the head sections of CAIs, as it involves the examination of a great variety of rings to which the chemical moiety, responsible for anchoring within the CA cavity, is attached. In addition, considerable efforts were also made toward the identification of new chemical entities able to bind the CAs. All these studies were aimed at both fulfilling the need of selectively targeting the desired CA isoforms as well as finding better chemical moieties for medicinal chemistry purposes. On recalling Fig. 9.2, the inhibition of the catalytic cycle from CAIs may occur mainly by means of (1) direct coordination of the inhibitor to the metal ion to form an adduct of the E-type or (2) by addition to the Zn(II) coordination sphere to afford the trigonal bipyramidal adduct F. This chapter will focus on the zinc-binding chemical moieties of CAIs, using the α -CAs as models, which encompass the two coordination geometries described above and discusses their features in detail. Some exceptions will also be reported.

9.3.1 Sulfonamides

Primary sulfonamides ($R\text{-SO}_2\text{NH}_2$) are the most widely used zinc-binding groups (ZBGs) among the CAIs and include the long-lasting clinically used CAI-based drugs (Fig. 9.5), since Mann and Keilin reported sulfanilamide in 1940 [17].

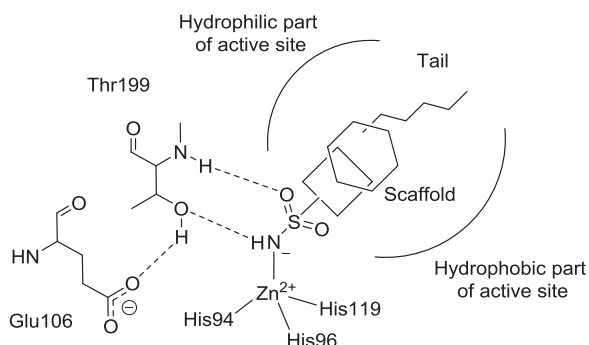
It is not surprising that primary sulfonamides have been investigated in depth by means of kinetic, physiological, and pharmacological studies [1,18]. More importantly, several

**Figure 9.5**

Chemical structures of some clinically used CAIs.

crystallographic contributions have been available, which cover many of the sulfonamide-based CAIs forming adducts with various CA isozymes [19–21]. Such a large amount of data represents a valid source for structure–activity relationship (SAR) analysis to elucidate the main structural factors responsible for the bindings, including the inner features. All primary sulfonamides anchor the CA isozymes according to the coordination cluster schematically reported in Fig. 9.6.

The sulfonamide moiety, which is deprotonated within the CA active site, binds to the catalytic Zn(II) ion after displacement of the metal-bound water molecule and according to a tetrahedral geometry. Additional stabilizations to the adduct are given by the primary NH and one oxygen of the sulfonamide, which interact with the Thr199 residue by means of hydrogen bonds. The Thr199 residue in turn is coordinated to the carboxylate moiety of

**Figure 9.6**

Schematic illustration of the key interactions between a primary sulfonamide and the hCA II active site as determined by X-ray crystallography [3].

Glu106 nearby, and this enhances acceptance of the hydrogen bonding ability of the Thr199 hydroxyl moiety [22,23]. Depending on the nature of the R scaffold mounted on the ZBG moiety, additional interactions with the hydrophilic and/or hydrophobic regions of the active site may occur and may further contribute to the stabilization of the whole adduct (Fig. 9.6).

Fig. 9.7 illustrates the binding modes of the phenyl (A), thiadiazole (B), and (C) thiadiazoline scaffolds decorated with the primary sulfonamide in adduct with the CA II.

The organic scaffolds in Fig. 9.7 are the prototypes of the well-known drug-based CAIs, sulfanilamide, acetazolamide, and methazolamide, respectively (Fig. 9.5). Therefore, no further examples of this kind are reported in this chapter. The reader is also directed to the appropriate chapters in this book where a selection of the most important, and/or promising, CAIs possessing the sulfonamide ZBG for medicinal chemistry purposes is reported. It is worth considering the contribution to the binding modes when an additional primary sulfonamide is inserted. A first example of this kind is represented by the nanomolar inhibitor CAI dichlorphenamide **4**, which has been largely clinically used for the management of glaucoma [24,25] and CNS-related disorders [1,19,26]. High-resolution X-ray crystallography of the **4**-hCA II adduct revealed the canonical interaction previously discussed, which anchors the molecule to the bottom of the enzyme cavity (Fig. 9.8A). The additional *m*-sulfonamide group was found facing toward the hydrophilic side of the CA enzymatic cleft and was reported to interact with Thr200, His64, Asn67, and Gln92 residues by means of hydrogen bonds [27].

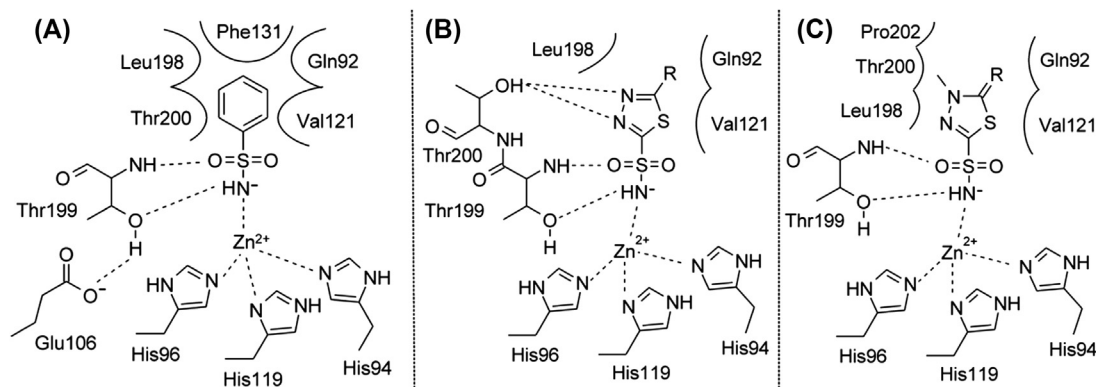


Figure 9.7

Schematic representation of the binding mode of phenyl (A), thiadiazole (B), and (C) thiadiazoline derivatives within the hCA II active site as determined by X-ray crystallography. Reprinted with permission from Alterio V, Di Fiore A, D'Ambrosio K, et al. Multiple binding modes of inhibitors to carbonic anhydrases: how to design specific drugs targeting 15 different isoforms? *Chem Rev* 2012;112:4421–68. Copyright 2012 American Chemical Society.

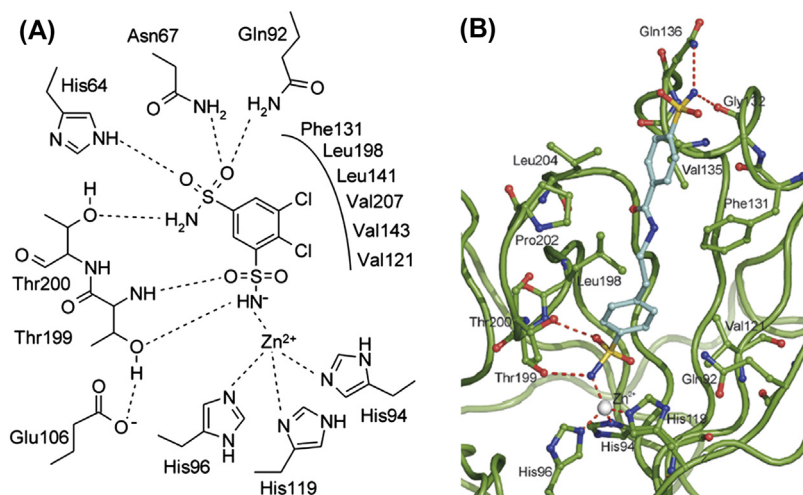


Figure 9.8

(A) Schematic representation of the binding mode of dichlorophenamide **4** within the hCA II active site (PDB accession code 2POU) [27]. (B) View of the active site region in the hCA II/**7** [4-sulfamoyl-N-(4-sulfamoylphenethyl)benzamide] complex showing residues participating in recognition of the inhibitor molecule. Reprinted with permission from Alterio V, Di Fiore A, D'Ambrosio K, et al. Multiple binding modes of inhibitors to carbonic anhydrases: how to design specific drugs targeting 15 different isoforms? *Chem Rev* 2012;112:4421–68. Copyright 2012 American Chemical Society.

Such interactions force the plane of the phenyl moiety to assume an orientation rotated by approximately 45 degrees and tilted by 10 degrees with respect to that generally observed in complexes of the type described above. Such findings were also reported for other bis-substituted sulfonamide-containing phenyl rings [3] and represent proof of the use of a CAI anchoring group (i.e., the primary sulfonamide) for obtaining proper orientation of the scaffold within the CA cavity. Such a concept was further developed with the insertion of two primary sulfonamide moieties at different phenyl rings properly spaced as in compound **7** [4-sulfamoyl-N-(4-sulfamoylphenethyl)benzamide, Fig. 9.8B] [3,28,29].

As reported above, the two sulfonamide moieties in **7** allow for optimal orientation of the CAI within the active site, with the first engaged in the canonical interactions with the enzymatic metal ion, whereas the terminal one was found to be participating in strong hydrogen bonds with the amino acid residues located at the rim of the active site (Fig. 9.8B) [28]. The structural findings relative to CAIs of the **4**- and **7**-type allowed for extension of the use of the primary sulfonamide groups within the field of CAIs. Such a moiety is generally reported in the literature as a zinc-binding group devoid of any structural feature that may determine orientation of the CAI scaffold within the enzymatic cleft. However, the data discussed above have shown that the sulfonamides, when properly

inserted within a molecular scaffold, contribute to the substrate orientation which makes them of consideration both for the design and development of future CAIs having selective inhibition profiles against the CAs of interest.

9.3.2 Bioisosteres of sulfonamides: sulfamates and sulfamides

Sulfamates and sulfamides ($R\text{--OSO}_2\text{NH}_2$ and $R\text{--NHSO}_2\text{NH}_2$, respectively) are properly defined as the bioisosteres of the sulfonamides previously described. The first reports on sulfamates as CAIs were from Maren [30] and Maryanoff [31]. More recently, in 1996, the simple sulfamide **8** and the sulfamic acid **9** bound to the metal ion of the Co(II)-substituted CA II were reported [32], whereas their X-ray structures in complex with native CA II were reported years later [23]. The binding modes of both **8** and **9** revealed a coordination pattern to the Zn(II) ion superimposable to the primary sulfonamide moiety. The adducts were also further stabilized by a hydrogen bond network within the enzymatic cavity as with the presence of additional nitrogen/oxygen atoms (Fig. 9.9) [23].

A large number of organic scaffolds bearing the sulfamate and sulfamide moieties have been reported in the literature. All were shown to possess highly effective inhibitory properties against the CAs [1] and some have relevant clinical applications, such as the fully protected β -D-fructopyranose sulfamate topiramate **10** which, as discussed in Chapter 15, is currently used for the treatment of epilepsy and obesity [7,16,18,33–35]. Within the scope of this chapter, the authors consider selected examples and focus on the ZBGs. The X-ray crystal structure of **10** in complex with hCA II was solved and showed that the sulfamate moiety directly coordinated to the Zn(II) ion according to a tetrahedral geometry (Fig. 9.10A). Again, the typical binding motif with Thr199 was present [37].

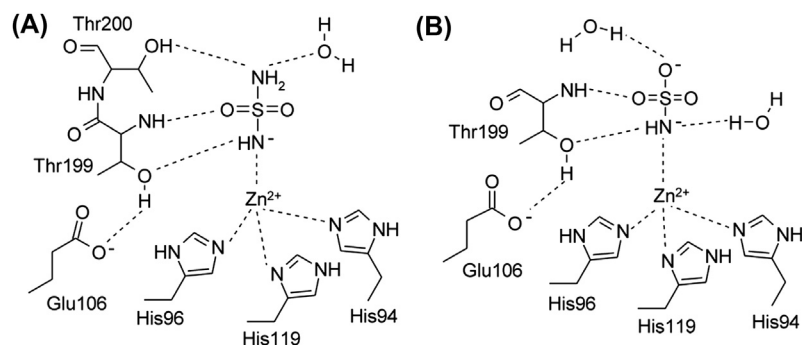


Figure 9.9

Schematic representation of the binding mode of (A) sulfamide **8** and (B) sulfamate **9** to the hCA II active site [23]. Reprinted with permission from Alterio V, Di Fiore A, D'Ambrosio K, et al. Multiple binding modes of inhibitors to carbonic anhydrases: how to design specific drugs targeting 15 different isoforms? *Chem Rev* 2012;112:4421–68. Copyright 2012 American Chemical Society.

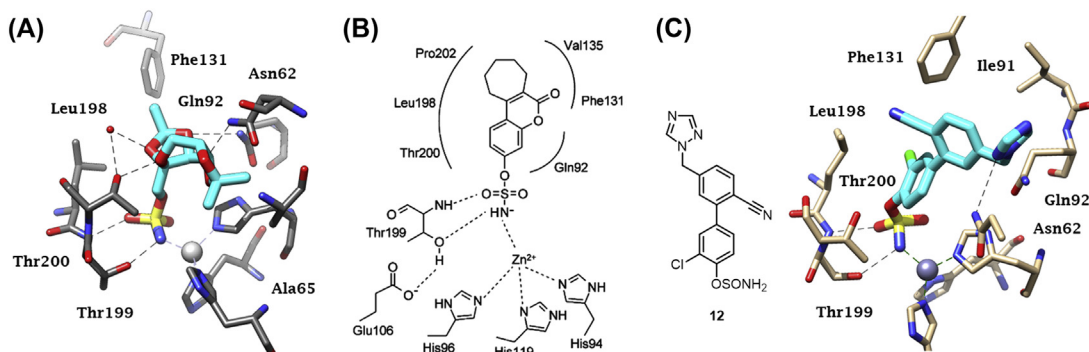


Figure 9.10

(A) Active site view of the hCA II/**10** adduct (PDB 3HKU). (B) Schematic view of the hCA II active site in complex with **11** (STX-64, PDB 1TTM). (C) Chemical structure of **12** and active site view of its adduct with hCA II (PDB 2WD3) [36]. (B) Reprinted with permission from Alterio V, Di Fiore A, D'Ambrosio K, et al. Multiple binding modes of inhibitors to carbonic anhydrases: how to design specific drugs targeting 15 different isoforms? *Chem Rev* 2012;112:4421–68. Copyright 2012 American Chemical Society.

Another valuable example of sulfamate-containing molecules is represented by inhibitors of the steroid sulfatase (STS; EC 3.1.6.2) enzymes. This class of compounds has shown quite good results for the treatment of hormone-dependent cancers (i.e., breast cancers), since they inhibit the production of estrogenic hormones which sustain the growth and development of this type of cancer [38]. The first-generation STS inhibitors (STSIs) presented a polycyclic organic scaffold decorated with a sulfamate moiety. Among them is the coumate STX-64 (**11**), also called oristusane [39,40], which was discontinued for the management of breast malignancies in 2018. The presence of the sulfamate moiety triggered structural investigations into such compounds with CAs (Fig. 9.10B) [36,41–47].

The binding of **11** within the CA II revealed the canonical tetrahedral fashion coordination of the Zn(II) ion [3]. The discontinuation of **11** opened the way for new structures to be explored as STSIs. An alternative approach for the treatment of hormone-dependent tumors consists of reducing levels of estrogens using aromatase inhibitors. Such molecules contain as an active pharmacophore a heme-chelating azole ring, such as triazole [43]. Since aromatase inhibitors given in conjunction with STSIs could maximize estrogen depletion in treated patients, new hybrid drugs able to inhibit simultaneously both enzymes have been designed. In particular, the X-ray crystal structure of the hCA II/**12** adduct was solved, showing a classical sulfamate–Zn(II) binding motif, the biphenyl moiety located at the center of the active site cavity, and the triazole ring pointing toward a small cavity positioned close to the entrance of the active site and delimited by Leu60, Asn62, and

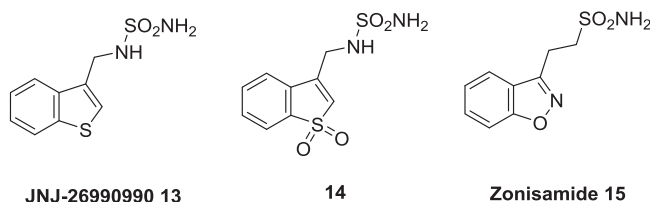


Figure 9.11

Chemical structures of JNJ-26990990 **13**, its *S,S*-dioxide derivative **14**, and zonisamide **15** [48].

Glu69 (Fig. 9.10C) [36]. All the contributions discussed above clearly indicate that the presence of a ZBG such as sulfamate may ascribe to the CAs a polypharmacological role in the treatment of many diseases.

Among the sulfamide-containing molecules, we report a recent study on JNJ-26990990 **13** and its *S,S*-dioxide derivative **14** as isostructural derivatives of the antiepileptic zonisamide **15** (Fig. 9.11) [48].

The inhibition data of the three compounds against the physiologically relevant human isoforms I, II, VII, IX, XII, and XIV demonstrated that sulfamides **13** and **14** were fairly similar to sulfonamide **15** (data not shown) [48]. The crystal structures of **13** and **14** in complex with the ubiquitous CA isoform II have been solved (Fig. 9.12) [48].

The initial |Fo–Fc| maps in both complexes clearly showed the binding of the inhibitor molecule at the active site region. This binding in both adducts does not induce any

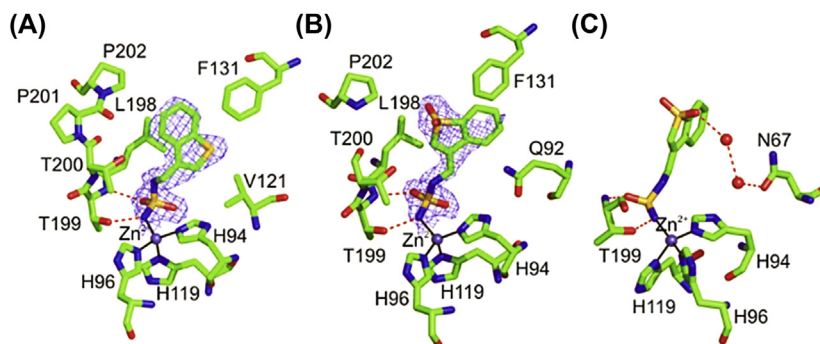


Figure 9.12

Active site region of the hCA II/**13** (A) and hCA II/**14** (B) adducts. The σ A-weighted $|2Fo - Fc|$ maps (contoured at 1.0 σ) relative to the inhibitor molecules are shown. The zinc ion coordination and residues involved in van der Waals interactions are also shown. The water-mediated hydrogen bond of the *S,S*-dioxide moiety of inhibitor **14** with Asn67 is depicted in (C). Reproduced from Di Fiore A, De Simone G, Alterio V, et al. The anticonvulsant sulfamide JNJ-26990990 and its *S,S*-dioxide analog strongly inhibit carbonic anhydrases: solution and X-ray crystallographic studies. *Org Biomol Chem* 2016;14:4853–8, with permission from The Royal Society of Chemistry.

significant change in the hCA II structure, as judged by the rmsd values of 0.26 and 0.27 Å calculated by superposition of all the C α atoms of the hCA II/**13** and hCA II/**14** adducts, respectively, with those of the native enzyme. Analysis of both structures reveals that inhibitors **13** and **14** maintain a binding mode to the CA active site similar to that observed for other CAIs containing a sulfonamide/sulfamide ZBG, which have been structurally characterized thus far [3].

9.3.3 Substituted sulfonamides

A class of less investigated CAIs is represented by the substituted sulfonamides. The first report of this kind was a contribution on saccharin **16** [49]: a sulfimide with a lactam and cyclic sulfonamide moiety. The latter functionality is responsible for the acidic character of the molecule, which also suggested a potential interaction with the CA enzymatic Zn(II) ion. An in vitro kinetic investigation of saccharin **16** against the human-expressed CAs was carried out and, surprisingly, some CA isoforms were strongly inhibited up to nanomolar concentrations (i.e., the VII and IX isoforms) [49]. Although saccharin **16** is a micromolar inhibitor of the CA II isoform, the corresponding adduct was obtained and investigated by means of X-ray crystallographic experiments [49].

Saccharin **16** is most likely coordinated in a deprotonated state through its nitrogen atom to the catalytically active zinc ion, thereby producing a flattened distorted tetrahedral environment at the metal center (Fig. 9.13A). The local environment around the nitrogen

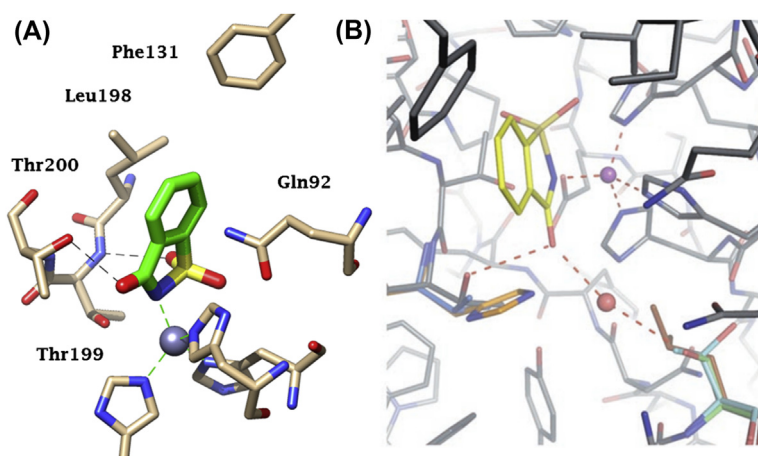


Figure 9.13

(A) Binding mode of saccharin **16** at the active site of hCA II (2Q1B). (B) Crystal structure from PDB 2Q1B (saccharin **16** in yellow, protein in gray) with superimposed residues mutated in the other isoforms: His200 (orange, CA I), Val200 (blue, mCA XIII), Ser65 (dark green, CAVII), Leu65 (brown, CA VA), and Thr65 (cyan, CA XIV) [49].

atom itself is pyramidal, with the hydroxyl group of the Thr199 side chain found along the assumed direction of the lone pair of electrons. The separation of 3.3 Å advocates for the formation of a weak hydrogen bond; however, the latter oxygen atom is also in hydrogen-bonding distance of one oxygen atom of Glu106 (2.5 Å). One oxygen atom of the neighboring sulfonyl group increases the coordination at the zinc center toward a distorted bipyramidal geometry (3.1 Å), while the second oxygen atom forms a weak hydrogen bond to the backbone NH group of Thr199. The lactam carbonyl oxygen atom is in hydrogen-bonding distance to the Thr200 hydroxyl group. Furthermore, it forms a hydrogen bond with a water molecule at the active site. Since saccharin **16** proved to be highly potent in inhibiting the CA VII (K_i 10 nM), a homology model retrieved from the Swiss-Model Repository [50] was used. Interestingly, an Ala residue is replaced by a Ser residue at the far end of the binding pocket in the vicinity of the lactam carbonyl oxygen atom (Fig. 9.13B) [49].

In the crystal structure of the saccharin **16**–CA II complex, a water molecule was observed which is ideally placed to mediate contact between the lactam carbonyl group and the serine hydroxyl group present in the homology model of CAVII (Fig. 9.13). This additional contact is likely to enhance the ligand binding [49]. In addition, sequence alignments showed that this Ala/Ser replacement also occurs in CA I, CA IV, CAVB, CA IX, CA XII, and murine CA XIII. CAIX and XII also showed enhanced saccharin binding, whereas a dramatic drop in affinity is observed for CA I and CA XIII. In the latter two isoforms Thr200 is replaced by either His200 or Val200, both of which require significantly more space next to the lactam carbonyl group. Thus, they are expected to produce steric conflicts with the bound saccharin, with a resultant dramatic drop in affinity. Weak inhibition is also observed in CA VA as the Ala to Ser replacement is exchanged by a spatially more demanding Leu residue. This residue would also be unable to promote water-mediated contact, and so, as with the Ala mutation, it will be detrimental for the potent binding of saccharin. CA VI and CA XIV are also inhibited to a remarkable extent by saccharin. Interestingly, they possess a Thr residue at the site of Ala65 in CA II. This residue should be equally well suited to pick up a water-mediated contact from the ligand [49].

An interesting study on open saccharin-based secondary sulfonamides has been recently reported and showed that compounds of the type reported in Fig. 9.14A were quite effective in inhibiting the tumor-associated isoforms IX and XII (data not shown) [51].

The binding mode of such compounds was determined by means of docking studies (Fig. 9.14B), which indicated that the hydroxymethyl group and one of the sulfonamide oxygen atoms could interact simultaneously with the Zn(II) ion, whereas the other sulfonamide oxygen atom was water accessible. This oxygen atom might also form hydrogen bonds with the backbone of Thr199 [51].

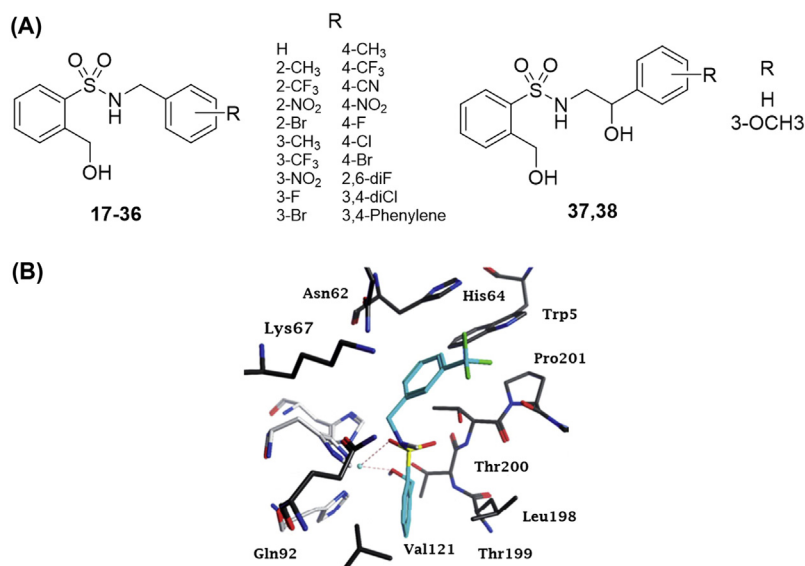


Figure 9.14

(A) Opened saccharin-based secondary sulfonamides **17–38** [51]. (B) In silico predicted binding mode of **23** at the active site of CA XII. Hydrogen bonds and interactions with the Zn(II) ion are depicted by red dashed lines. The Zn(II) ion is depicted as a turquoise sphere [51].

9.3.4 Thiol derivatives

The validation of the thiol moiety as an effective CAI was achieved by means of studies devoted to decipher the CA inhibition mechanism of the biologically active compound thioxolone **39**. Interestingly the X-ray crystallographic data on the hCA II/**39** adduct revealed that the 4-mercaptobenzene-1,3-diol molecule **40** bound to the catalytic zinc ion was present instead (Fig. 9.15) [53].

The CAI **40** was directly coordinated to the zinc(II) ion through the thiol moiety and to the Thr199 oxygen atom by means of a weak hydrogen-bond interaction. Interestingly the binding mode of the thiol moiety **40** was closely related to the previously discussed sulfonamides and their bioisosteres [3]. The phenyl ring was stabilized by means of several van der Waals interactions, while the hydroxyl group in *para*-position was shown facing toward a hydrophobic enzymatic pocket delimited by the Ile91, Phe131, Val135, Leu141, and Leu198 residues and established a series of nonfavorable interactions. The CAI **40** was determined as the real species inhibiting the CA enzyme (K_i of 148 μ M in vitro), and a mechanistic investigation into thioxolone **39** was carried out and a plausible pathway hypothesized (Fig. 9.15).

According to the mechanism proposed, thioxolone **39** undergoes CA-mediated hydrolysis in two steps resulting in H₂CO₃ and thiol **40** as the final products. The mechanism proposed was further corroborated by a kinetic investigation of the single chemical species considered

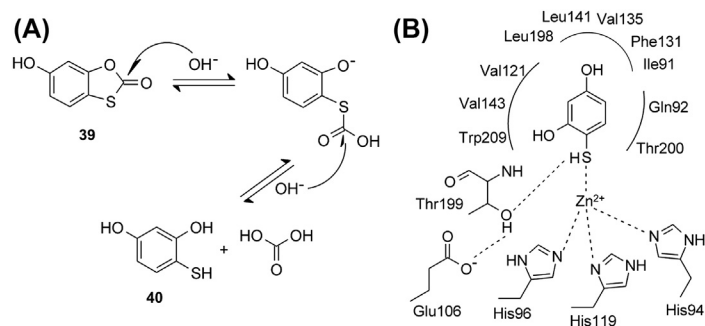


Figure 9.15

(A) Proposed CA II-mediated hydrolysis of thioxolone **39** [52]. (B) Schematic representation of the active site region in the CA II/**40** complex (PDB 2OSF). Reprinted with permission from Alterio V, Di Fiore A, D'Ambrosio K, et al. Multiple binding modes of inhibitors to carbonic anhydrases: how to design specific drugs targeting 15 different isoforms? *Chem Rev* 2012;112:4421–68. Copyright 2012 American Chemical Society.

above (e.g., **39** and **40**). This study proved that the mercaptophenols are a new class of CAI [53,54]; however, proper use of such compounds as CAI-based drugs still needs to be investigated.

9.3.5 Selenols

The different atomic structure and composition of selenium compared with sulfur makes the selenol (SeH) group of SeCys (pKa 5.8) more nucleophilic than the thiol (SH) group of Cys (pKa 8.3). Selenols with a low pKa value [55] under physiological conditions (pH 7.4) were proved to be almost always present as selenolate species (R-Se^-), whereas the majority of thiols are present in their nonionized form (R-SH). Taking into account the soft character of the selenium atom, the polarity of the Se–H bond, and the affinity of selenium compounds for zinc finger proteins [56], selenols were investigated as a new chemotype as human (h) CAIs. The in vitro kinetic data of a series of selenols were tested against CA I, II, VII, and IX, which revealed their relatively strong inhibition. Then the binding mode of selenols was investigated by means of X-ray experiments on compound **41** in complex with CAs I and II at 1.64 and 1.44 Å resolutions, respectively (Fig. 9.16A and B) [57,58].

The crystallographic data revealed the binding mode of selenols to be quite different with respect to the typical binding mode of phenols. As discussed in Chapter 10, phenols were found anchored to the nonprotein zinc ligand by a hydrogen bond involving the H atom of the inhibitor and the donor OH from the zinc hydroxide [58]. In contrast, selenol was bound directly to the Zn(II) ion of the CA active site with the usual tetrahedral geometry, and similar to the binding observed for thiolates, as discussed in the previous section [59]. In particular, the selenol **41** scaffold was found to make hydrophobic

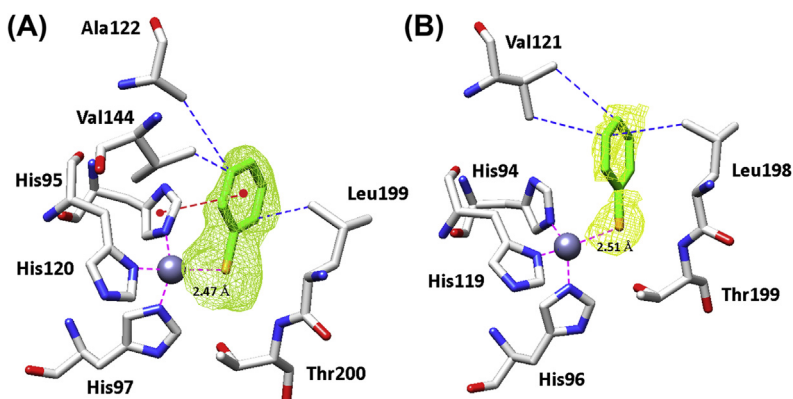


Figure 9.16

(A) Active site region of the CA I/**41** adduct (PDB 6HWZ). Inhibitor shown as a σ_A -weighted $|F_o - F_c|$ density map at 2.0σ . (B) Active site region of the hCA II/**41** adduct (PDB 6HX5). Inhibitor shown as a σ_A -weighted $|F_o - F_c|$ density map at 2.0σ . van der Waals interactions, π -stacking interactions, and the active site Zn(II) coordination are also shown and labeled in blue, magenta, and red, respectively. Reproduced from Angeli A, Tanini D, Nocentini A, et al. *Selenols: a new class of carbonic anhydrase inhibitors*. *Chem Commun (Camb)* 2019;55:648–51, with permission from The Royal Society of Chemistry.

interactions with Ala122, Val144, and Leu199 residues within the hCA I active site. Furthermore, the imidazole ring of His95 was also involved in a π -stacking interaction with the aromatic ring of the inhibitor (Fig. 9.16A). The crystal structures of hCA II (Fig. 9.16B) also revealed a similar binding mode for selenol **41**, which was observed to be directly bound to the Zn(II) ion. However, the electron density relative to the inhibitor was less clear, indicative of a less ordered position of the inhibitor when compared to the hCA I complex. This might be explained by the fact that **41** is bound less effectively to this isoform, since a limited number of interactions among the ligand and the amino acid residues occurred (i.e., **41** showed hydrophobic interactions only with Val122 and Leu198).

9.3.6 Benzoxaboroles

Experimental evidence that benzoxaborole moiety may be an efficient CAI was first reported in 2016. In this pioneering study, a small series of benzoxaborole-based compounds was investigated in vitro for their ability to inhibit the most relevant hCAs (e.g., I, II, IX, and XII) and the data obtained clearly showed some activity, which was significantly influenced by the tail moieties decorating the main scaffold (Fig. 9.17A) [60].

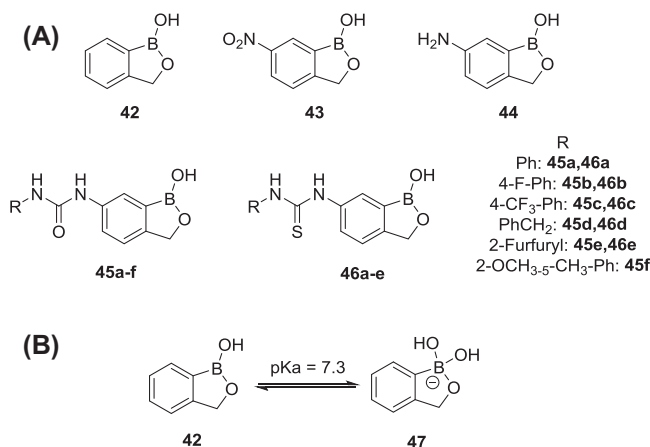


Figure 9.17

(A) Chemical structures of benzoxaborole derivatives [60]. (B) Lewis/Brønsted acidic properties of benzoxaboroles [60].

The high-resolution crystal structure of the hCA II/42 adduct was solved, and inspection of the electron density maps at various stages of the crystallographic refinement revealed the binding of the inhibitor to the enzyme active site in its tetrahedral anionic form **47** (Fig. 9.17B). This finding is in agreement with the observation that at the pH used for crystallization experiments (i.e., 8.7), compound **42** is mainly present in solution in this form [61]. Interestingly, two different binding modes, both characterized by the coordination of the tetrahedral derivative to the catalytic zinc ion, were observed (Fig. 9.18) [60].

In the first, hereafter indicated as binding mode **A**, one of the hydroxyl groups of benzoxaborole **42** was anchored to the catalytic zinc ion, completing its tetrahedral coordination (Fig. 9.18A), whereas, in the second, hereafter indicated as binding mode **B**, the inhibitor was bound to the zinc ion with two of its oxygen atoms, generating a trigonal bipyramidal coordination geometry of zinc (Fig. 9.18B). In both binding modes, additional polar and van der Waals interactions contributed to the stabilization of the adduct. The observation that in both cases binding of the inhibitor to the enzyme was mediated by the tetrahedral anionic form **47** prompted us to investigate which was the binding mode of **42** at the physiological pH, where, on the basis of the equilibrium reported in Fig. 9.17B, the trigonal planar form of the inhibitor was also present. To clarify this point, new crystallization conditions of the hCA II/42 adduct at pH 7.4 were set. Interestingly, the analysis of the obtained structure revealed that also in this case the inhibitor was bound to the enzyme in its anionic tetrahedral form **6**, adopting two binding modes identical to those observed in the structure crystallized at pH 8.7. Based on these findings, the binding mode of benzoxaborole **42** to the enzyme active site was revealed to

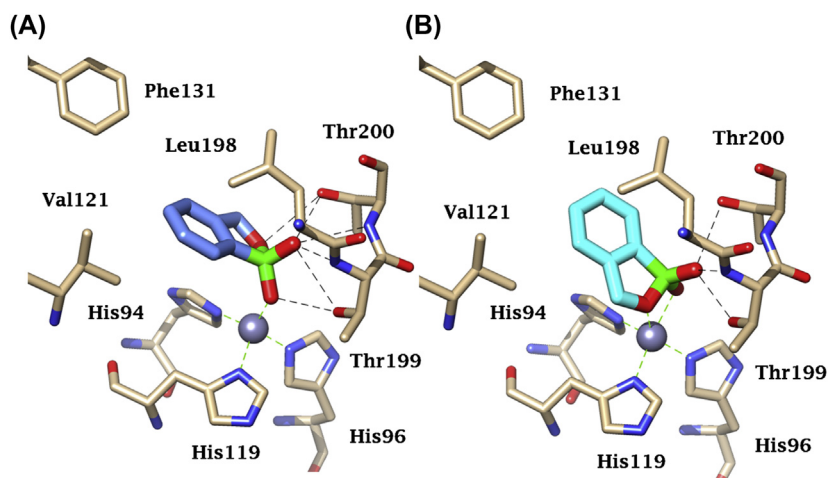


Figure 9.18

Active site view of the hCA II/42 adduct crystallized at pH 8.7. The two possible coordination modes of the inhibitor are depicted. (A) Binding mode A: tetrahedral coordination. (B) Binding mode B: trigonal bipyramidal coordination [60].

be independent of pH and always associated with the anionic tetrahedral form **47**, probably because in this way the electron avid boron atom has eight electrons in its last shell. The observation that the tetrahedral form is involved in the mechanism of action of the benzoxaboroles is in agreement with previously reported data on benzoxaboroles as inhibitors of leucyl-tRNA synthetase [62,63].

9.3.7 Phosphates and phosphonates

Phosphates and phosphonates are important motifs found in a large variety of molecules of biological interest, especially anticancer or antiviral drugs. The use of the phosphonate motif in the design of CAIs was first reported in 2000 [64]. A small series of substituted aryl *N*-phosphoryl sulfonamides [$-\text{SO}_2\text{NHPO}(\text{OH})_2$] were prepared and studied kinetically for inhibition of reactions catalyzed by CA I and II (Fig. 9.19A).

Some derivatives exhibited a strong inhibitory activity against CA II, with an inhibition constant below 20 nM (data not shown) and some were potent CAIs as their corresponding parent unsubstituted sulfonamides. By analogy, sulfamates possessing the phosphoryl moiety ($\text{R}-\text{O}-\text{SO}_2\text{NH}-\text{PO}_3\text{H}_2$) have also been reported and demonstrated that the phosphorylated sulfamate zinc-binding group was very efficient for the design of low-nanomolar CAIs [66]. A detailed kinetic inhibition study of five CAs with inorganic phosphates **54** and **55** (PO_4^{3-} , HPO_4^{2-}), carbamoyl phosphate **56** ($\text{H}_2\text{NCOOPO}_3^{2-}$),



(A) Structures of phosphoryl sulfonamides **48a-i–53** [64]. (B) Schematic representation of the interaction between the foscarnet **57** and the hCA I active site [65].

It is worth pointing out that compared to all other CA inhibitor–adducts investigated to date by means of X-ray crystallography, no strong hydrogen bond was evidenced between the OH group of Thr199 and the inhibitor atom directly coordinated to the Zn(II) ion. Moreover, a repulsive interaction between one methyl group of the side chain of Leu198 and the oxygen atom of the carboxylate moiety of foscarnet **57** was observed. This type of clash was already observed for other CAIs and may explain the relatively weak inhibitory activity of foscarnet **57** against hCA I [68]. Further kinetic studies were reported in the literature on this kind of moiety and all demonstrate that in addition to the classical sulfonamide motif, the phosphonate function has a strong affinity for the active site of different carbonic anhydrase isozymes and can be included in the class of zinc-binding function that can be used in the design of selective CAIs [69].

9.3.8 Benzenephosphonamidates

A recent contribution reported the phenylphosphonic diamide **58** and its alkyl phosphonamidate derivatives to inhibit CA I, II, IV, VII, and IX (Fig. 9.20). The in vitro obtained data showed that the simplest and less sterically hindered compounds were the most potent inhibitors against all the evaluated isoforms with K_i s in the low micromolar range [70].

In silico studies were performed in order to determine the binding modes of compounds within the catalytic sites of CAs [70]. It is worth noting that phenylphosphonic diamide **58** assumed a sulfonamide-like binding mode with the negatively charged nitrogen coordinating the Zn(II) ion and the NH donating an H-bond to the Thr199 side chain hydroxyl group. Moreover, the P=O moiety accepted an H-bond from the same residue backbone NH. The other NH₂ of **58** slightly protrudes into the internal pocket lined by Val143, Trp9, Val121, and Leu198. The aromatic ring is π -alkyl stacked by Val121 and Leu198 side chains (Fig. 9.21A).

Not surprisingly, two alternative and clearly distinct binding modes were found for the substituted phosphonamidates (only **59–61** are reported as selected examples), according to the *R/S*-configuration of the ZBG central atom. The stabilization of a unique binding mode for enantiomer was driven by the common necessity to bind the Zn ion through the NH moiety and to establish a proper network of H-bonds with the residues nearby. As a result, the *R*-enantiomers assumed an orientation whereby the coordinating NH protrudes toward the target pocket and the P=O is in H-bond distance with Thr199, thus exposing the alkyl groups to the pocket defined by Ala65, Asn244, and His64 (Fig. 9.21B). In a completely opposite manner, the *S*-enantiomers maintain mostly the same set of interactions, except for the alkyl moieties that occupy the target “deep water” area lined by the hydrophobic Val121, Trp9, Leu198, and Val143 side chains (Fig. 9.21C) [70].

The in silico data reported above were also integrated with pK_a studies in order to better define the binding modes of the phosphonamidates, as some free-energy discrepancies were observed when compared with analogous binders such as the sulfonamides. This is the

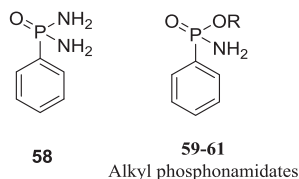


Figure 9.20

Chemical structure of phenylphosphonic diamide **58** and general structure of alkyl phosphonamidates **59–61** [70].

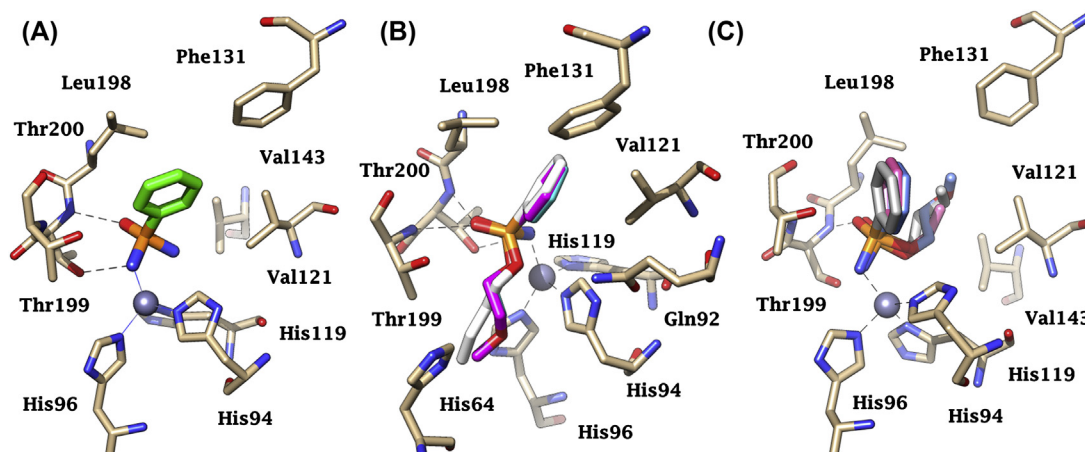


Figure 9.21

QPLD/MM-GBSA predicted poses of the studied compounds with hCA II: (A) **58** (green); (B) **59**-(*R*) (light blue), **60**-(*R*) (magenta), and **61**-(*R*) (white); and (C) **59**-(*S*) (blue), **60**-(*S*) (purple), and **61**-(*S*) (grey) [70].

preliminary study providing evidence that phosphonamidates can be efficiently CAIs, presenting interesting inhibition properties depending on the chiral configuration.

9.3.9 Inorganic anions

Inorganic anions are weak CAIs with affinities in the millimolar range, and only a few have shown low micromolar inhibition potencies against CAs (e.g., cyanide **62**, azide **63**, hydrogen sulfide **64**, and trithiocarbonate **65**) [1,4,71,72]. Interestingly the X-ray crystallographic investigation on the obtained adducts with CA II has shown that anions, not only the inorganic ones, may bind within the enzymatic CA site either by direct coordination of the metal ion or by occupation of a secondary cavity in close proximity to the Zn(II) ion [1,4,71,73,74]. In this chapter, only the metal coordinating anions will be considered, which have been reported to assume three different coordination patterns: (1) regular tetrahedral [5,74]; (2) slightly distorted tetrahedral [72,75,76], and (3) trigonal bipyramidal [5,77]. As a general rule, the anionic species able to act as hydrogen-bonding donors toward the active site residue Thr199 are reported to directly coordinate the Zn(II) ion according to a regular tetrahedral geometry after displacement of the bound water/hydroxide ion. Examples of this kind are hydrogen sulfide **64** (HS^-) and hydrogen bisulfite **66** (HSO_3^-) (Fig. 9.22A) [5,23,71,74].

Ligand species devoid of protons may adopt either a distorted tetrahedral or trigonal bipyramidal coordination geometry. Examples include the azide **63** (N_3^-) [75,76] and the bromide **67** (Br^-) [75], which both coordinate to the Zn(II) ion after displacement of the

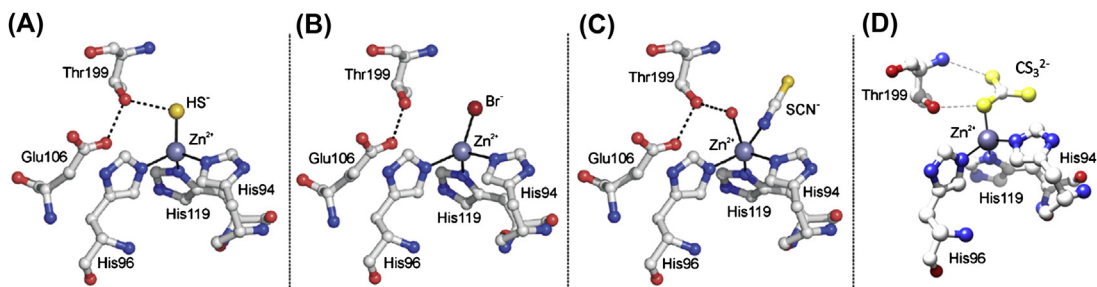


Figure 9.22

Representation of the coordination geometries of (A) hydrogen sulfide **64**, (B) Br^- **67**, (C) thiocyanate **69**, and (D) trithiocarbonate **70** (CS_3^{2-}) within the hCA II active site (PDB 1CAO, 1RAZ, 2CA2, and 3K7K) [74]. Reprinted with permission from Alterio V, Di Fiore A, D'Ambrosio K, et al. Multiple binding modes of inhibitors to carbonic anhydrases: how to design specific drugs targeting 15 different isoforms? *Chem Rev* 2012;112:4421–68. Copyright 2012 American Chemical Society.

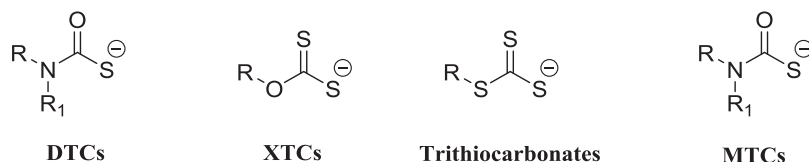
metal-bound water/hydroxide ion. Since these inhibitors cannot interact with the Thr199 residue, as usually occurs for CAI metal binders, the tetrahedral geometry was found to be significantly distorted (Fig. 9.22B) [75,76].

Other examples of the binding mode for the anions as CAIs are represented by the formate **68** [5] and the thiocyanate **69** [78], which were found bound to the enzyme by addition to the metal coordination sphere to afford an adduct with a trigonal bipyramid geometry (Fig. 9.22C) [5,78].

Although the ions suffer when used directly in medicinal chemistry, within the field of CAIs, they assume a high value as represented in the blueprint for the design and development of new binding species as in the case of the low micromolar CAI trithiocarbonate **70** (CS_3^{2-}) ion (Fig. 9.22D) [72,79–81]. The X-ray crystal structure for the adduct of CS_3^{2-} **70** bound to CA II showed the anion coordinated to the enzymatic Zn(II) ion by means of one of the sulfur atoms in a slightly distorted tetrahedral geometry, and occupying a position similar to that observed in the case of the hCA II–bicarbonate complex [72]. The same sulfur made a hydrogen bond to the OH of Thr199, whereas a second sulfur atom participated with another hydrogen bond to the NH group of the same amino acid residue. On the basis of the trithiocarbonate **70** (CS_3^{2-}) data, the dithiocarbamates ($\text{RR}_1\text{N-CS}_2^-$; DTCs) containing moieties were developed as CAI-organic surrogates of the inorganic ion [79,80].

9.3.10 Dithiocarbamates (DTCs), xanthates (XTCs), and monothiocarbamates (MTCs)

The use of DTC-bearing compounds in large amounts began in the early 20th century, with their industrial application as accelerators in the rubber vulcanization process [82].

**Figure 9.23**

Chemical structures of DTCs, XTCs, trithiocarbonates and MTCs.

Such a method proved to be far more efficient as compared with the primitive sulfur-based procedures introduced by Goodyear and Hancock approximately 60 years earlier [82]. **DTCs** (Fig. 9.23) have been established in different fields such as in analytical, macromolecular and organic chemistry, medicine, and biology. Undoubtedly, plant disease management is the area in which **DTCs** are most dominant [82]. The role of **DTCs** will strengthen in the near future, due to both global population growth (about nine billion in 2050) and the necessity to control plant diseases by means of chemical approaches [82,83]. The main feature of **DTCs** is their ability to form stable complexes with metal and alkali-metal ions, and many reports accounting for the characterization of such structures are available in the literature [82]. However, the first study dealing with **DTCs** and metalloenzymes only appeared in 1983 with the investigation of *N,N*-diethyldithiocarbamate on bovine (b) CAs [84].

The known metal-chelating properties of the **DTCs** and the preliminary investigations previously reported set the starting point to validate this moiety as an efficient CAIs class. In this context a series of DTC-containing organic scaffolds were synthesized and kinetically profiled against human- and bacterial-expressed CAs and in general very high enzymatic affinities were found (K_i s in the low nanomolar range) [79]. Selected **DTCs** were particularly efficient in lowering the intraocular pressure in an in vivo model of glaucoma [79]. More importantly in the context of this chapter, investigations were carried out to define the binding modes of selected **DTCs** **71–73** by means of X-ray crystallographic experiments of the corresponding adducts with CAs (Fig. 9.24) [79,80].

Data collection showed DTCs deeply buried in the enzyme active site. The CS_2^- groups, upon displacement of the zinc-bound water/hydroxide ion, directly interacted through a sulfur atom with the catalytic zinc ion at a Zn–S distance of 2.3 Å. The angles defined by the closely associated zinc-coordinating His (His94, His96, and His119), the metal ion, and the **DTCs** sulfur atom accounted for a distorted tetrahedral geometry, which resembled the binding mode of the classical CAIs, such as the sulfonamides and their bioisosteres, sulfamates and sulfamides (Fig. 9.24A–C) [79,80]. The major differences in the interaction modes of **71–73** with the hCA occur at the organic scaffold level. The benzyl ring of compound **73** engages a T-shaped π -stacking interaction with the imidazole ring of His64, which, as a result, was frozen in the “in” conformation. Furthermore, **73** is stabilized by the

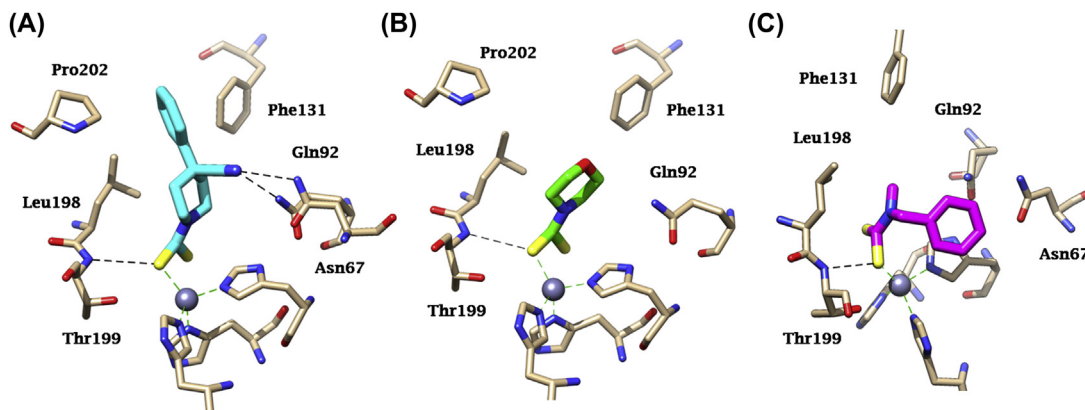


Figure 9.24

Binding modes of **DTCs** **71**–**73** within the hCA II active site [79,80].

weaker van der Waals interactions occurring with the side chains of Asn62, Glu92, His94, Val121, Phe131, Leu198, Thr200, and Pro202 (Fig. 9.24C) [79,80]. On the other hand, compound **71** positioned its phenyl ring onto a hydrophobic pocket located at the opposite site of the enzymatic tunnel bearing the His64 residue (Fig. 9.24A) [79,80]. The pendant His64 imidazole ring was a distance of 5 Å from **71**, and thus was not involved in any valuable interaction with the inhibitor and was allowed to flip adopting both the “in” and “out” conformations. The phenyl ring of **71** headed toward the rim of the catalytic cleft, interacting with Phe131 and Pro202, whereas the cyano group hydrogen bonds with Asn67 and Gln92 and a water molecule locate nearby (Fig. 9.24A) [79,80]. Compound **72** was the deepest buried of them all, and, for **71**, no interactions with the His64 side chain occurred. The only contacts detected for the compound **72** scaffold in complex with CA II were with two water molecules, Wat393 and 540 (Fig. 9.24B) [79,80].

Replacement of the nitrogen atom in **DTCs** by an oxygen atom resulted in the formation of xanthate moieties as depicted in Fig. 9.23 (ROCS₂[−]; **XTCs**) [85].

In analogy to the **DTCs** discussed previously, various series of **XTCs** were prepared and tested in vitro for their inhibition potencies against the human-expressed CAs (Fig. 9.23) [85]. The kinetic data accounted for low nanomolar *K_is*, and for the **XTCs** a potent IOP-lowering activity in vivo was proved also [85]. Although no **XTC**–CA adducts were obtained, probably due to the high liability of the moiety itself, the binding modes of such compounds were determined by means of in silico calculations using an analogous **DTC** derivative as a template [85]. Interestingly, the results obtained showed **XTCs** coordinated monodentately to the Zn(II) ion of the enzymatic active site, almost in a superimposable

manner with the **DTCs** for which the X-ray crystal structure was reported earlier [80]. The Zn–S distances were 2.12–2.46 Å for **XTCs** and 2.31 Å for the **DTC**. The sulfur atoms of these **XTC/DTC** were also engaged in hydrogen bonds with the side chain hydroxyl of Thr199 (with calculated distances of 3.0–3.5 Å for **XTCs** and 3.1 Å for the **DTC**). The organic scaffolds of the evaluated **XTCs** adopted extended conformation, engaging in favorable contacts within the enzyme active site, similar to the binding of the **DTC** [80]. All these interactions are in agreement with the excellent kinetic in vitro data measured against the CA II isoform (K_i s of 5.4–8.1 nM).

An additional compound analog to the above-discussed **DTCs** and **XTCs** is the organic benzyl trithiocarbonate **74** (BnSCS_2^-), which was obtained by means of isosteric substitution of the oxygen atom with a sulfur atom (Fig. 9.23) [85]. The in vitro kinetic data on the human-expressed CA isoforms are the only available data so far. Interestingly, such a compound showed potent low nanomolar inhibition potency against the tumor-associated CA isoforms IX and XII. The binding mode of **74** has not been determined yet either experimentally or by means of in silico calculations, but it is reasonable to assume it to be closely matched with the progenitor **DTCs** and **XTCs**.

The bioisosteric variation was also explored within the coordinating moiety, as for the monothiocarbamates (**MTCs**; $\text{R}_1\text{RN-COS}^-$). In such a case, a sulfur atom was substituted with an oxygen atom, as illustrated in Fig. 9.23 [86]. The in vitro kinetic data of the **MTCs** obtained showed interesting inhibition properties against the physiologically relevant hCAs. In general, hCA I was poorly inhibited, followed by the catalytically more efficient hCA II, and then by tumor-associated hCA isoforms hCA IX and XII, which were strongly inhibited by **MTCs** with K_i values in the low nanomolar range [86]. A selection of water-soluble **MTCs** was tested in vivo for their IOP-lowering properties which were shown to be far superior to the reference and clinically used dorzolamide at the same concentration [86].

To elucidate the binding mode of the **MTC** moiety within the CA active site, molecular docking studies were performed on compound **75** (N-phenylmonothiocarbamate), which was demonstrated to have good inhibitory properties against hCAs II, IX, and XII. Docking of **75** into the X-ray crystal structure of hCA II (PDB 3P5L) predicted that the ligand binds deep in the enzyme catalytic site in its anionic form, with the **MTC** moiety chelating the Zn(II) ion in a monodentate manner (Fig. 9.25A) [86].

In this regard, calculations suggested that the metal ion can be alternatively coordinated through the sulfur or oxygen atom. However, previous spectroscopic and crystallographic studies indicated that **MTCs** coordinate metal ions through sulfur because the resulting negative charge distribution mainly shifted toward this atom [87]. Thus, the lowest energy docking pose among those showing the sulfur–Zn(II) coordination was considered.

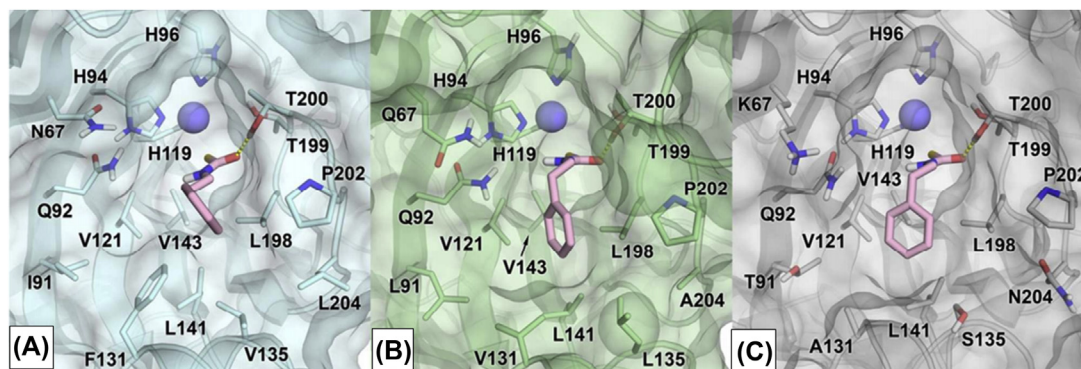


Figure 9.25

Binding mode of compound **75** within the active site of (A) CA II (PDB 3P5L), (B) CA IX (PDB 3IAI), and (C) hCA XII (PDB 1JD0). The ligand is shown as pink sticks. Reprinted with permission from Vullo D, Durante M, Di Leva FS, et al. *Monothiocarbamates strongly inhibit carbonic anhydrases in vitro and possess intraocular pressure lowering activity in an animal model of glaucoma. J Med Chem* 2016;59:5857–67. Copyright 2016 American Chemical Society.

As reported above, the **MTC** oxygen atom in compound **75** interacted through a hydrogen bond with the T200 residue and in analogy with many zinc-binding CAIs [88]. The phenylethyl moiety was extended from the enzymatic cavity toward the hydrophobic region of the CA-binding pocket, establishing a T-shaped stacking interaction with F131 and favorable van der Waals contacts with F131, V135, L141, L198, and P202. It is noteworthy that this binding pose was highly superimposable with the **DTCs** in complex with hCA II of comparable potency [80]. Docking calculations also helped to elucidate the molecular bases for the slight selectivity profile of the **MTC 75** derivative toward the tumor-associated hCA IX and XII. Multiple sequence alignment of hCAs I, II, IX, and XII showed that single-point mutations occur at the binding cavities of these enzymes. Specifically, the bulky F131 residue in hCA II was replaced by the smaller valine (F131V) and alanine (F131A) in hCA IX and XII, respectively. Interestingly, docking of compound **75** into hCA IX and hCA XII (Figs. 9.25B and C) revealed that these substitutions, although causing the loss of the highly stabilizing T-shaped π – π interaction, allow the ligand to penetrate deeper into the hydrophobic region of the active site, thus enhancing the contribution of desolvation to the overall ligand-binding free energy.

The inactivity of **MTC 75**, and more generally the lower activity of **MTCs**, against the hCA I compared to the other three isoforms may be rationalized as in both CA I N67 and T200 amino acids are replaced by a His residue, thus affecting the ligand binding for steric reasons. Moreover, the T200H substitution results in the loss of the key hydrogen bond established by the ligand **MTC** moiety in hCA II, IX, and XII. The following SARs

were obtained: (1) in the case of N-alkyl disubstituted **MTCs** showing long linear chains, they may not be hosted in the enzyme cavity for steric reasons, explaining why compounds featuring an *n*-butyl chain were inactive toward the CA II enzyme; (2) hydrophilic compounds may compensate for the loss of lipophilic contacts by establishing an additional hydrogen bond with the Q92 side chain; and (3) for aromatic substituted derivatives, a short chain linking the ring and the **MTC** moiety or meta substitutions on the terminal aryl ring should hamper the interactions with the residues lining the hydrophobic pocket (i.e., F131).

9.3.11 Benzyl carbamates

Few studies on the inhibition of CAs by carbamates have been reported in the literature [89–92]. In particular, in 1992 Parr and Khalifah measured the inhibition of the esterase activity of hCA I and hCA II by a series of *N*-unsubstituted carbamates [91], showing that these molecules bind to the CA active site through their deprotonate carbamate nitrogen, which coordinated the catalytic metal ion. These authors also demonstrated that these compounds show slow binding kinetics, due to their high pK_a, which is unfavorable for their deprotonation when interacting with the enzyme at physiological pH values [91,92]. More recently, the synthesis and inhibition properties of some N-monosubstituted carbamates have been reported [89,90]. Also in this case, the inhibition of the esterase activity of hCA I and hCA II was measured, proving that these compounds effectively inhibit hCAs at the nanomolar level. A recent investigation into carbamates as CAIs has provided a detailed inhibition and structural study of a very simple representative of this class of compounds, namely benzyl carbamate **76** (Fig. 9.26).

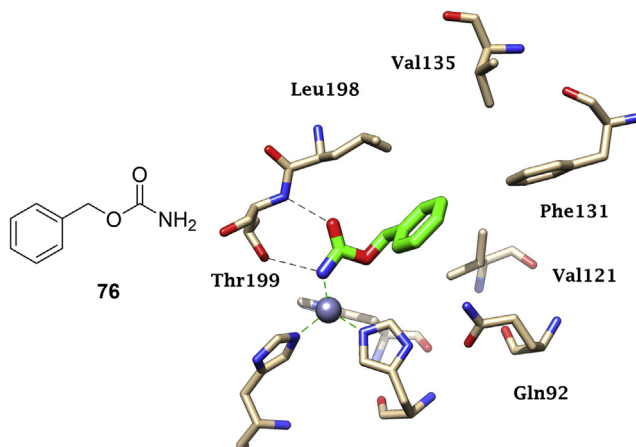


Figure 9.26

Structure of benzyl carbamate **76** and active site view of its adduct with hCA II (PDB 6H29).

Compound **76** was tested in vitro for inhibition of the CO₂ hydration activity of several physiologically relevant hCA isoforms [93] and its inhibition properties were compared to the simple benzenesulfonamide. When compound **76** was preincubated with various CA isoforms for 15 min, no detectable inhibition was observed (data not shown), in agreement with the previously described slow binding kinetics. When benzyl carbamate was incubated for 6 h with various CA isoforms, low micromolar range K_I values were obtained for the CA II and XII isoforms (data not shown).

To elucidate the binding mode of unsubstituted carbamates to the CA active site, the crystallographic structure of the adduct that compound **76** forms with hCA II was determined. As shown in Fig. 9.26, the inhibitor coordinated to the Zn(II) ion in a tetrahedral geometry by means of its nitrogen atom, which substitutes the water molecule/hydroxide ion present in the free enzyme. In agreement with previously reported spectroscopic studies on a cobalt-substituted CA, it was assumed that nitrogen is in its deprotonated form [91].

The inhibitor amide nitrogen is also involved in a strong hydrogen bond with the Thr199 hydroxyl group, whereas the carbonyl oxygen accepts a hydrogen bond from the main chain nitrogen of the same residue. No other polar interactions are present between the enzyme and the inhibitor. The benzyl moiety is located in the middle of the active site, establishing several hydrophobic interactions with residues Val121, Phe131, Leu141, Val143, and Leu198.

Interestingly, the ZBG of compound **76** is almost superimposable on bicarbonate bound to CA II. Although the difference between the pK_a of carbonic acid and this carbamate is huge, the metal ion promotes the deprotonation of the nitrogen NH₂ of **76**, and its coordination to the zinc ion is in a very similar manner to bicarbonate. In fact, the two oxygen atoms from the carbamate moiety and the corresponding oxygen from bicarbonate adopt superimposable orientations in the two complexes. Thus, the carbamates may be considered as substrate analogs.

Taken together, these results clearly indicate that N-unsubstituted carbamates can represent interesting lead compounds, as alternatives to the classical sulfonamides, to further explore the development of selective CAIs. Due to the lower affinity of the carbamate group for the CA active site, it is conceivable that the selectivity of such derivatives toward the different CA isoforms can be easily modulated by suitable substituents of this ZBG. Moreover, the peculiar orientation of the benzyl moiety of benzyl carbamate **76** within the CA active site can also be utilized to modulate the selectivity by means of suitable derivatization of the ring. This study represents the first structural evidence by means of X-ray crystallography of the CA–carbamate interaction, and of the direct binding of these compounds to a metal ion in a biological system.

9.3.12 Ureates and hydroxamates

CA II is not only a very effective catalyst for the interconversion between carbon dioxide and bicarbonate, but also shows some catalytic versatility, participating in several other hydrolytic processes which presumably involve nonphysiological substrates. These reactions include the hydration of cyanate to carbamic acid [94] or of cyanamide to urea [95,96], the aldehyde hydration to gem-diols [97,98], the hydrolysis of some carboxylic [99,100] or sulfonic acid esters [101], as well as other less investigated hydrolytic processes including the thioesters [102]. Among these reactions, the hydration of cyanamide to urea has been carefully characterized by means of spectroscopic, kinetic, and crystallographic techniques [95,96]. These studies demonstrated that cyanamide acts as a weak inhibitor of the esterase activity of CA II, interacting with the zinc ion within the enzyme active site. Interestingly, when CA II crystals were soaked with cyanamide for 90 min, two different adducts with different occupancies were observed in the active site. In its high occupancy form, the cyanamide replaces the hydroxide ion coordinating to the zinc ion with a regular tetrahedral geometry. In its low occupancy form, cyanamide adds to the coordination sphere [96] generating a pentacoordinated adduct together with a zinc-bound water molecule, which can be considered as a zinc outer-sphere fifth ligand (Fig. 9.27A). While the first adduct was

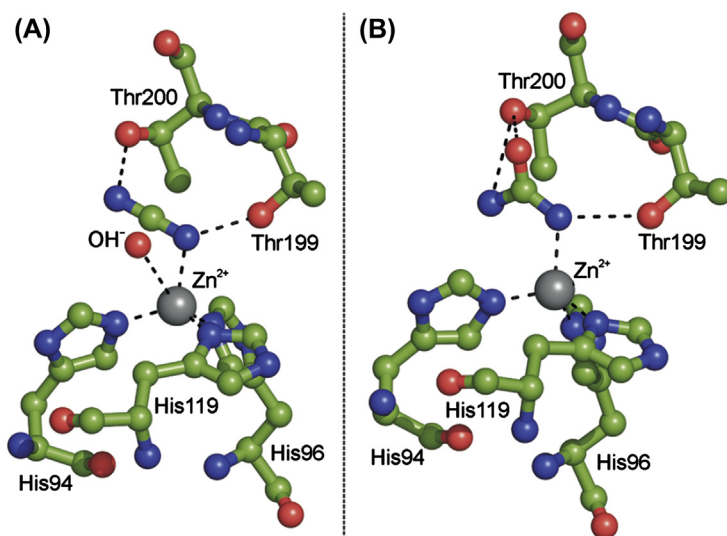


Figure 9.27

Representation of the binding mode of (A) cyanamide (PDB 1F2W) (pentacoordinated adduct), and (B) urea (PDB 1BV3) to the hCA II active site. Reprinted with permission from Alterio V, Di Fiore A, D'Ambrosio K, et al. Multiple binding modes of inhibitors to carbonic anhydrases: how to design specific drugs targeting 15 different isoforms? *Chem Rev* 2012;112:4421–68. Copyright 2012 American Chemical Society.

considered a dead-end product, the second one represented an intermediate of the catalyzed reaction, where the hydroxide ion was ready to perform the nucleophilic attack on the substrate cyanamide forming urea. In agreement with this hypothesis, when the CA II crystals were soaked in cyanamide solutions for several hours, the reaction took place with the resulting urea which remained bound to the metal ion (Fig. 9.27B) [95].

The observation of a strong binding between urea and the zinc ion has opened the way to new studies on the interactions of CAs with small molecules possessing appropriate structural features to act as zinc-binding functions. Among these molecules, N-hydroxyurea **77** represents one of the simplest compounds [103]. The crystallographic structure of the hCA II/N-hydroxyurea **77** adduct showed that this molecule binds to the Zn(II) ion of the hCA II active site in a bidentate mode, by means of the oxygen and nitrogen atoms of the NHOH moiety. Additional hydrogen bonds involving the hydroxyl and carbonyl moieties of the inhibitor and the enzyme residue Thr199 were also observed (Fig. 9.28A) [104]. It is worth noting that the related acetohydroxamic acid **78** and trifluoroacetohydroxamic acid **79**, although containing the same hydroxamate functionality, were demonstrated to adopt a very different binding mode. In particular, in the case of acetohydroxamic acid **78**, the ionized nitrogen was directly bound to the Zn(II) ion in a tetrahedral coordination (Fig. 9.28B), while in the case of trifluoroacetohydroxamic acid **79**, a fluorine atom and the ionized nitrogen formed a five-membered chelate with the zinc ion, with a coordination geometry that was approximately trigonal bipyramidal (Fig. 9.28C) [105]. Altogether, these findings suggest that the hydroxamate moiety represents a very versatile zinc-binding function which can adopt different binding modes to the metal ion present in the CA active site. Therefore, it could be usefully exploited for obtaining potent CA inhibitors.

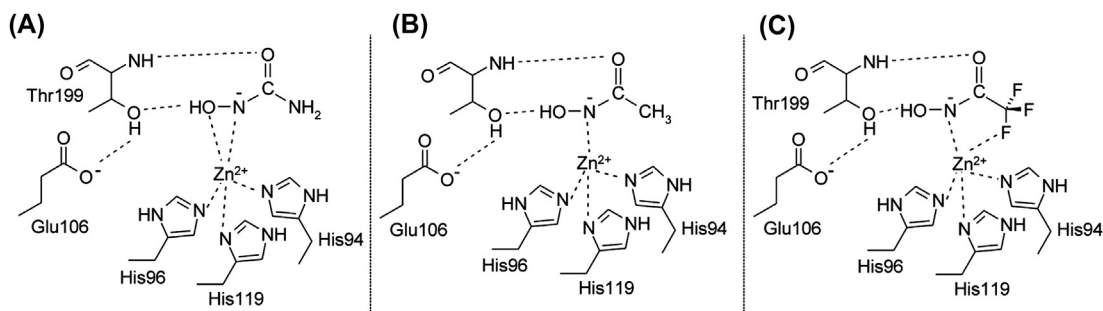


Figure 9.28

Schematic representation of (A) N-hydroxyurea **77** (PDB 2GEH), (B) acetohydroxamic acid **78** (PDB 1AM6), and (C) trifluoroacetohydroxamic acid **79** to the hCA II active site. *Reprinted with permission from Alterio V, Di Fiore A, D'Ambrosio K, et al. Multiple binding modes of inhibitors to carbonic anhydrases: how to design specific drugs targeting 15 different isoforms? Chem Rev 2012;112:4421–68.*

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9.4 Conclusions

This chapter reports selected examples of CAIs which exert their role by means of coordination to the catalytic Zn(II) ion deeply located at the bottom side of the CA cavity, and the main structural features have been discussed. In addition to the recurrent theme on metal coordination, the variety of chemical moieties considered is of interest. This implies that a plethora of chemical structures of various origins are exploitable for medicinal chemistry, biotechnological, or any CA-based purposes.

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